

Problem Set 4

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1 Weak and Strong Induction

Problem 1: Suppose that the only things you know about the multiplication symbol are that (for all naturals m and n)

$$0 * n = 0 \quad \text{and} \quad (1 + m) * n = n + m * n$$

Prove that

$$\forall n \in \mathbb{N}, n * 0 = 0$$

Problem 2: Prove the following using induction for all natural numbers m and n (feel free to know standard properties of addition and multiplication):

$$\sum_{i=1}^n m = n * m$$

Problem 3: Prove that player 2 *always* wins the simple version of the matches game (where each player takes one match from the pile of their choice), regardless of their strategy. (Note that proving this directly using induction might leave you with too weak of inductive hypothesis. Consider proving a *stronger* claim by induction and then concluding that this statement is true.)